Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

PPG12 Analysis Team (automated report)

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1 Introduction

In *pp* collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples (`photon10_double`, `jet12_double`) are blended with nominal single-interaction MC at the physics-dictated fractions: 18.7% double interaction at 0 mrad crossing angle, 7.2% at 1.5 mrad.

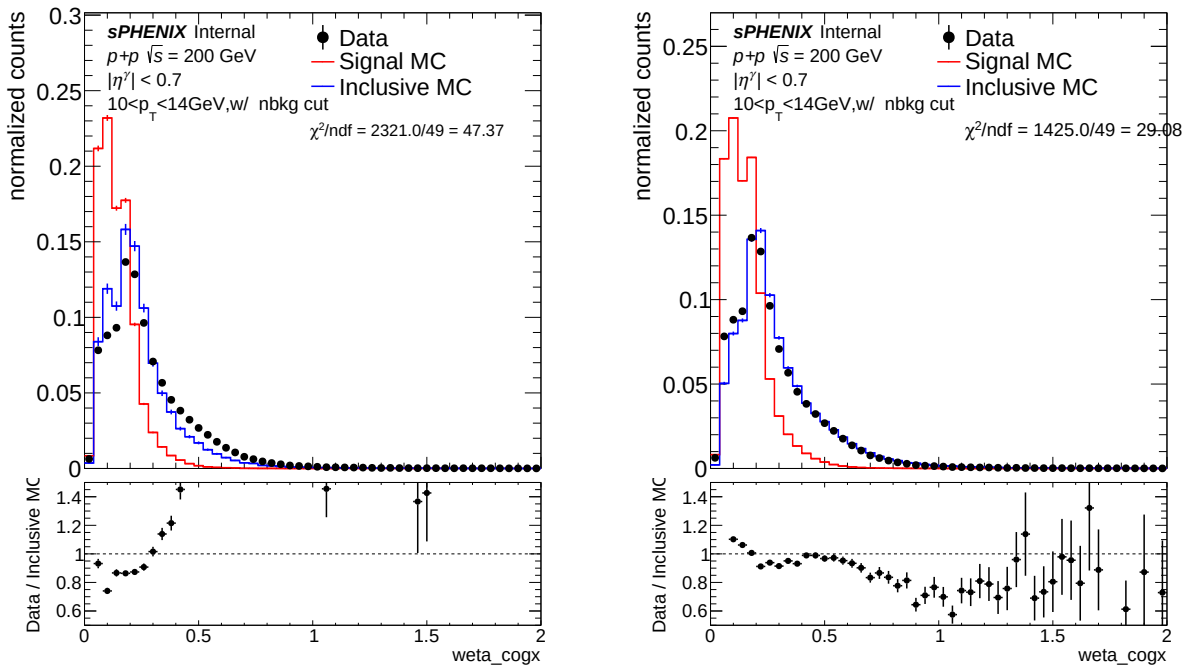
The blending uses a two-pass pipeline: Pass 1 builds a combined vertex distribution (weighted `hadd` of single and double vtxscan files); Pass 2 reruns the full shower shape analysis with consistent vertex reweighting and `mix_weight` applied to all histogram fills. The final "mixed MC" is the `hadd` of single (weight 0.813) and double (weight 0.187) output files.

This report compares data–MC agreement for four representative shower shape variables at the **cut1** selection level (with NPB preselection) for the **0 mrad** crossing-angle period. Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with χ^2/ndf and a Data/Inclusive MC ratio panel. Left panels show the nominal (single-only) MC; right panels show the mixed (double-blended) MC.

2 Shower Shape Comparisons

2.1 w_η^{CoG} (Eta Cluster Width)

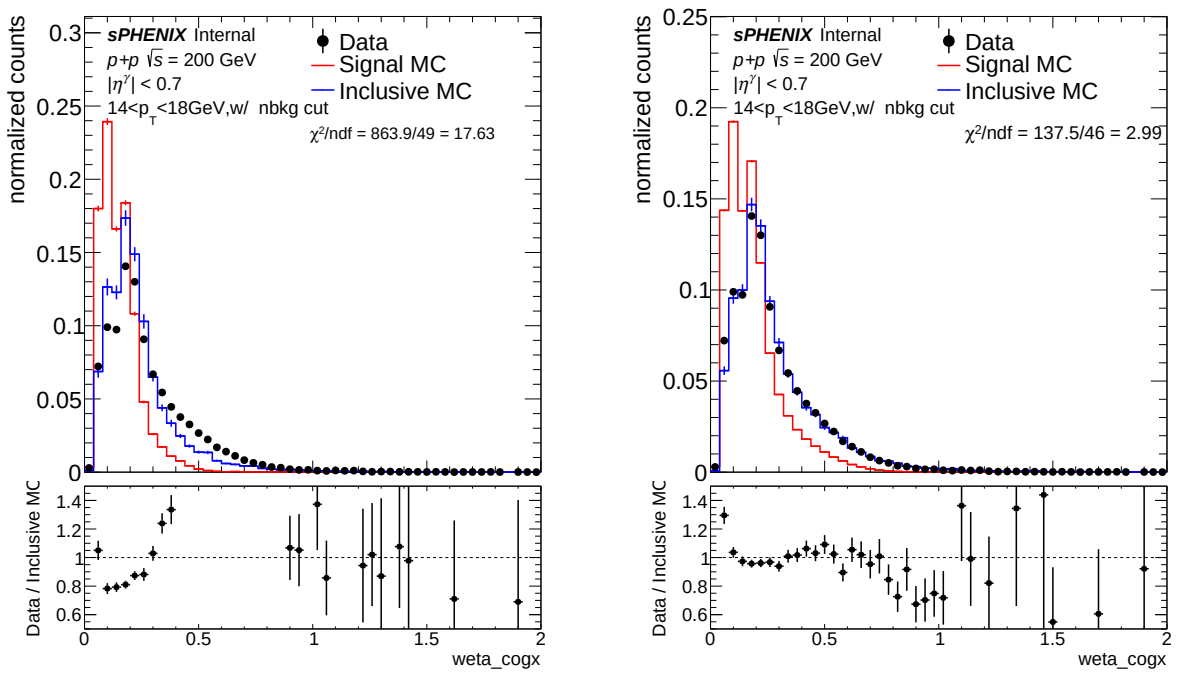
w_η^{CoG} is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile.



(a) Nominal MC

(b) Mixed MC

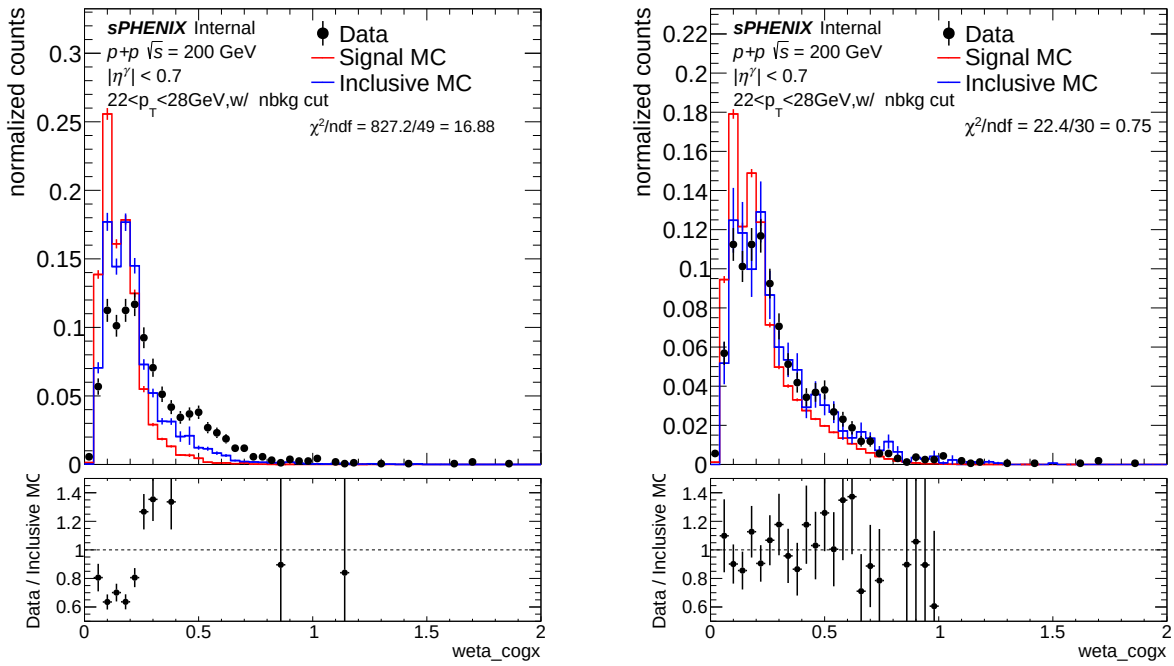
Figure 1: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0). Nominal $\chi^2/\mathrm{ndf} = 47.37$; mixed $\chi^2/\mathrm{ndf} = 29.08$.



(a) Nominal MC

(b) Mixed MC

Figure 2: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1). Nominal $\chi^2/\mathrm{ndf} = 17.63$; mixed $\chi^2/\mathrm{ndf} = 2.99$.



(a) Nominal MC

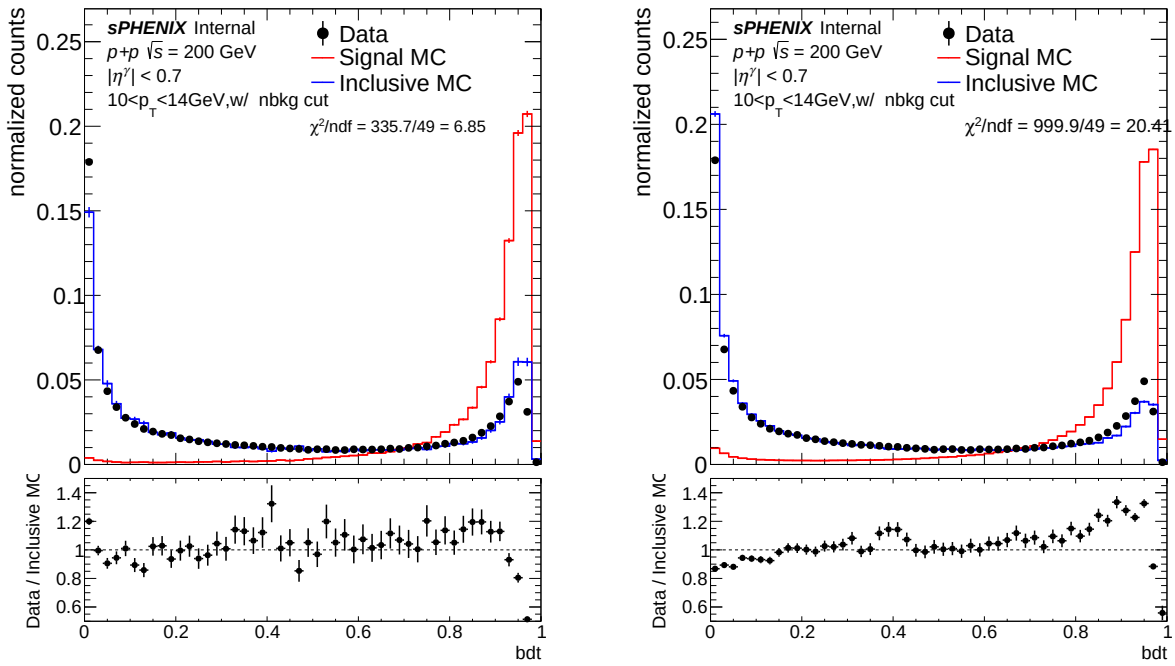
(b) Mixed MC

Figure 3: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3). Nominal $\chi^2/\mathrm{ndf} = 16.88$; mixed $\chi^2/\mathrm{ndf} = 0.75$.

The mixed MC dramatically improves data–MC agreement for w_η^{CoG} across all E_T bins, with χ^2/ndf dropping by factors of 1.6 (pt0) to 22 (pt3).

2.2 BDT Score

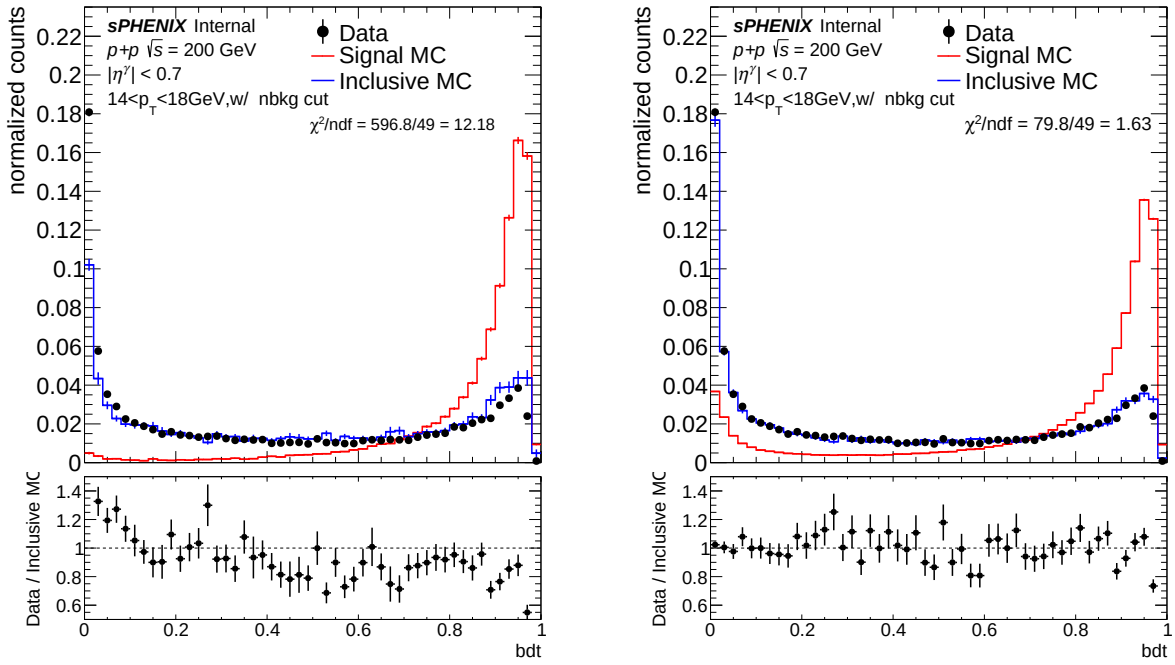
The photon/jet BDT discriminator is the most sensitive variable to pileup effects, as it amplifies underlying shower shape distortions through the XGBoost decision function.



(a) Nominal MC

(b) Mixed MC

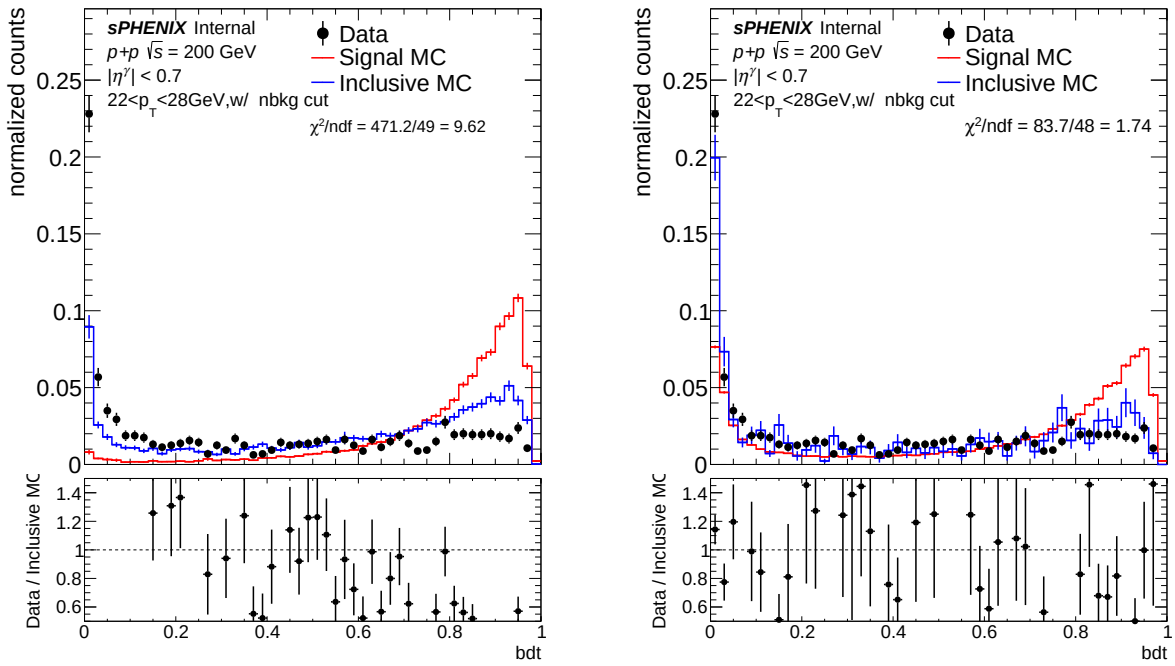
Figure 4: BDT score, $10 < E_T < 14\,\mathrm{GeV}$ (pt0).



(a) Nominal MC

(b) Mixed MC

Figure 5: BDT score, $14 < E_T < 18\,\mathrm{GeV}$ (pt1). Nominal $\chi^2/\mathrm{ndf} = 12.18$; mixed $\chi^2/\mathrm{ndf} = 1.63$.



(a) Nominal MC

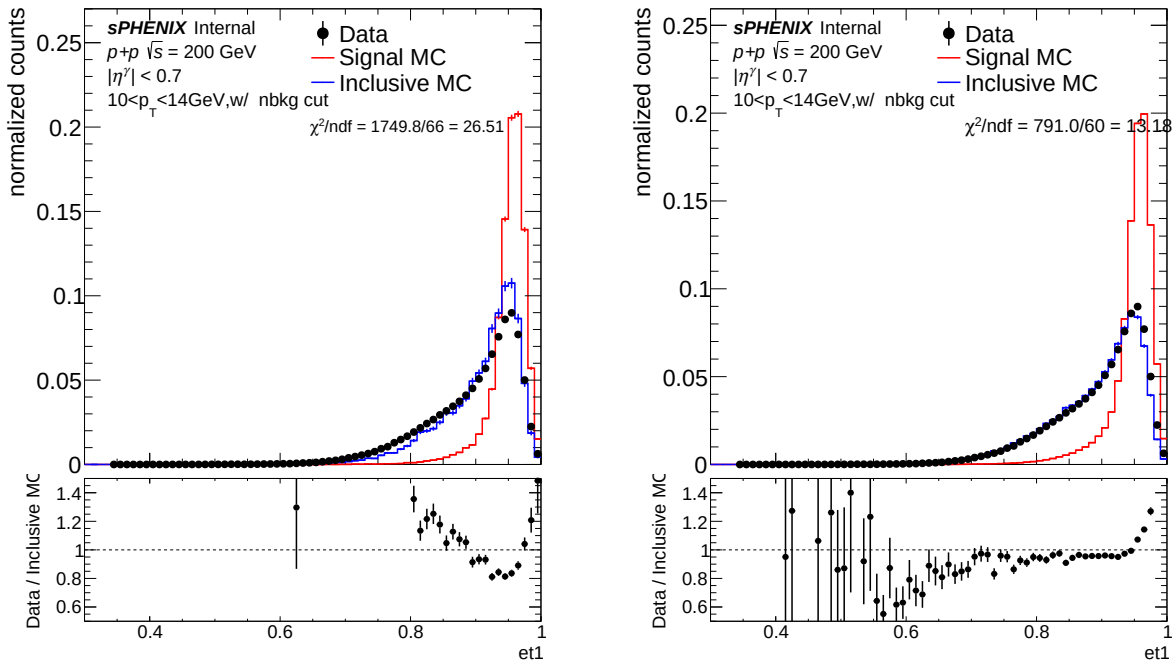
(b) Mixed MC

Figure 6: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3).

At pt1, the BDT χ^2/ndf drops from 12.18 (nominal) to 1.63 (mixed) — a factor of 7.5 improvement, confirming that the double-interaction admixture accounts for much of the nominal data–MC BDT discrepancy.

2.3 f_1 (Inner Ring Energy Fraction)

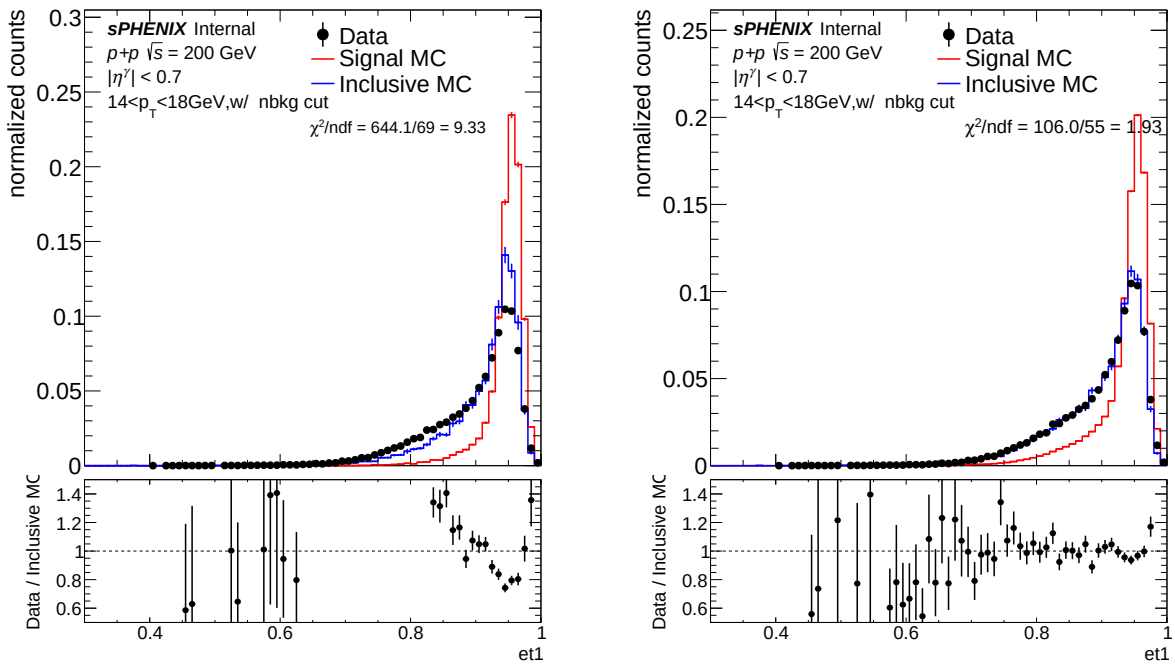
f_1 measures the energy fraction in the innermost ring around the seed tower. Double interactions spread energy outward, reducing f_1.



(a) Nominal MC

(b) Mixed MC

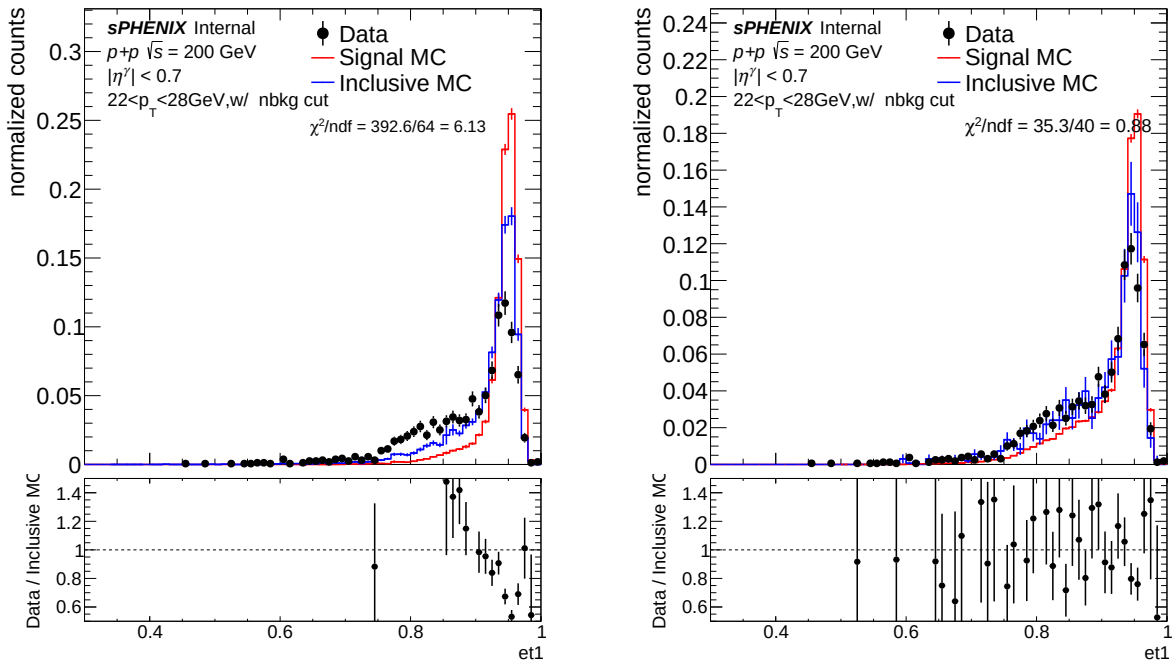
Figure 7: f_1, $10 < E_T < 14\,\mathrm{GeV}$ (pt0).



(a) Nominal MC

(b) Mixed MC

Figure 8: f_1, $14 < E_T < 18\,\mathrm{GeV}$ (pt1). Nominal $\chi^2/\mathrm{ndf} = 9.33$; mixed $\chi^2/\mathrm{ndf} = 1.93$.



(a) Nominal MC

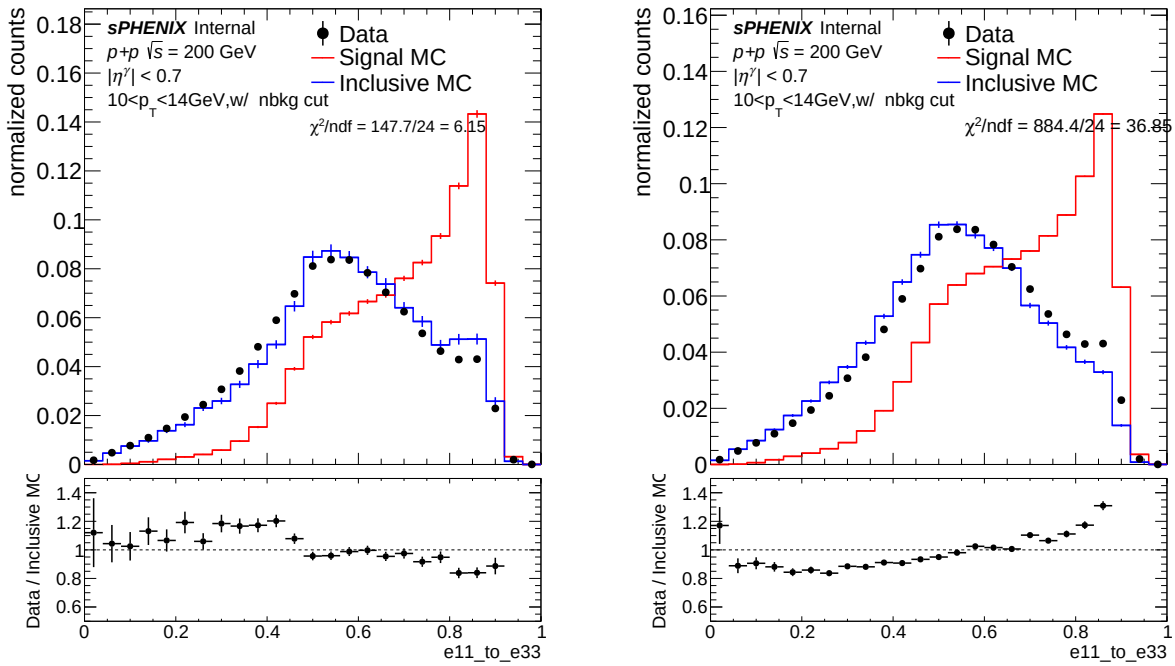
(b) Mixed MC

Figure 9: f_1, $22 < E_T < 28\,\text{GeV}$ (pt3).

At pt1, f_1 shows a factor of 4.8 improvement (χ^2/ndf from 9.33 to 1.93).

2.4 $E_{1\times1}/E_{3\times3}$ (Seed/Core Ratio)

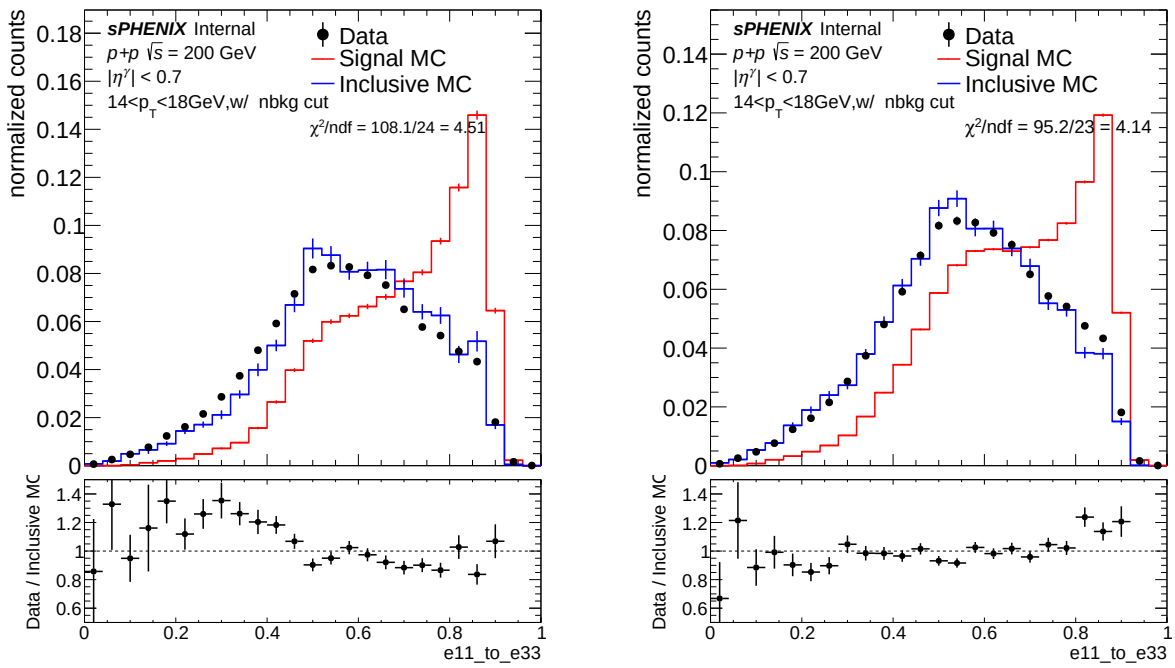
The seed-to-core energy ratio is sensitive to shower broadening: as energy migrates from the seed tower to neighbors, the ratio decreases.



(a) Nominal MC

(b) Mixed MC

Figure 10: $E_{1\times1}/E_{3\times3}$, $10 < E_T < 14\,\text{GeV}$ (pt0).



(a) Nominal MC

(b) Mixed MC

Figure 11: $E_{1\times1}/E_{3\times3}$, $14 < E_T < 18\,\text{GeV}$ (pt1). Nominal $\chi^2/\text{ndf} = 4.51$; mixed $\chi^2/\text{ndf} = 4.14$.

$E_{1\times1}/E_{3\times3}$ shows only a modest improvement (χ^2/ndf from 4.51 to 4.14 at pt1), indicating this variable is less sensitive to the double-interaction admixture than w_η^{CoG} or the BDT score.

3 Single vs Double Interaction Overlay

The plots in this section overlay single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized on the same canvas. These directly visualize where data falls relative to the two MC interaction types.

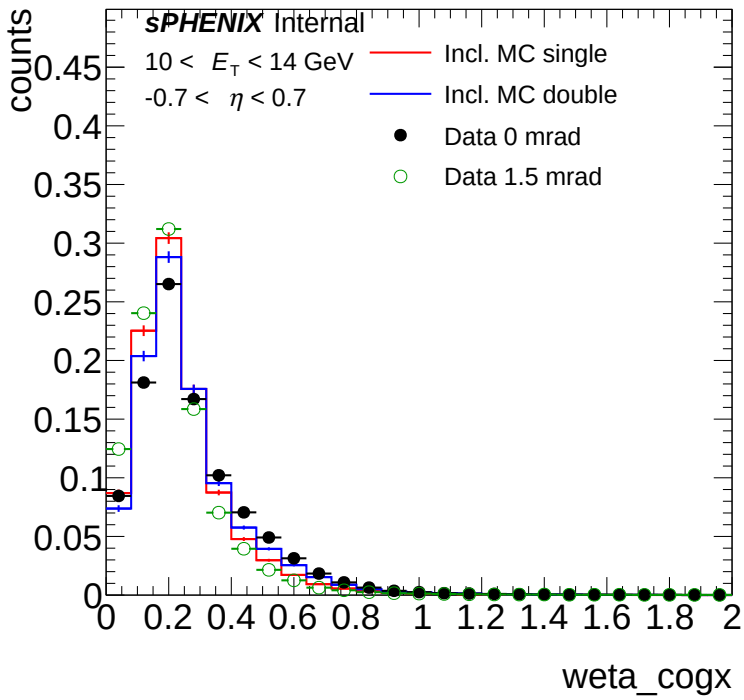


Figure 12: w_η^{CoG}, $10 < E_T < 14$ GeV (pt0). Data 0 mrad (black) falls between single (red) and double (blue) MC, while data 1.5 mrad (green) tracks closer to single MC.

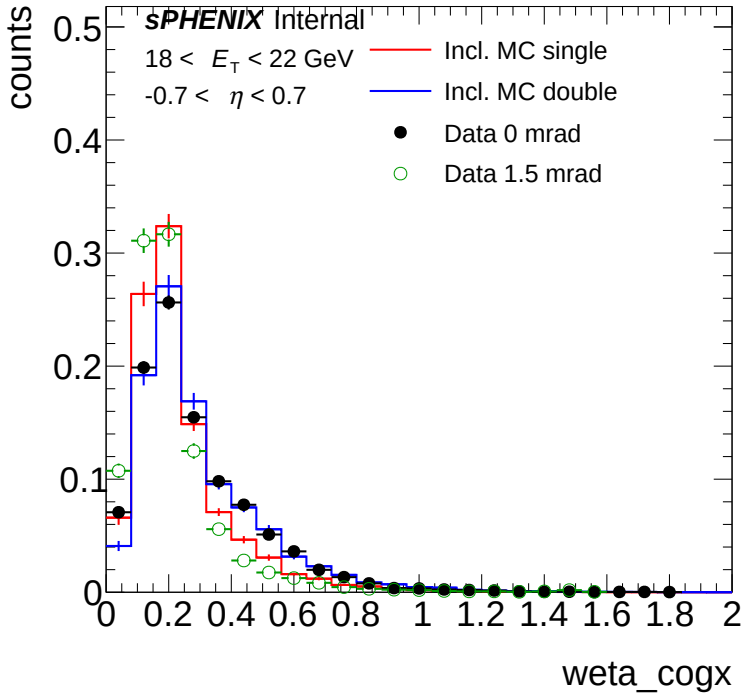


Figure 13: w_η^{CoG}, $18 < E_T < 22\,\mathrm{GeV}$ (pt2). The separation between single and double MC grows with E_T; data remains between the two.

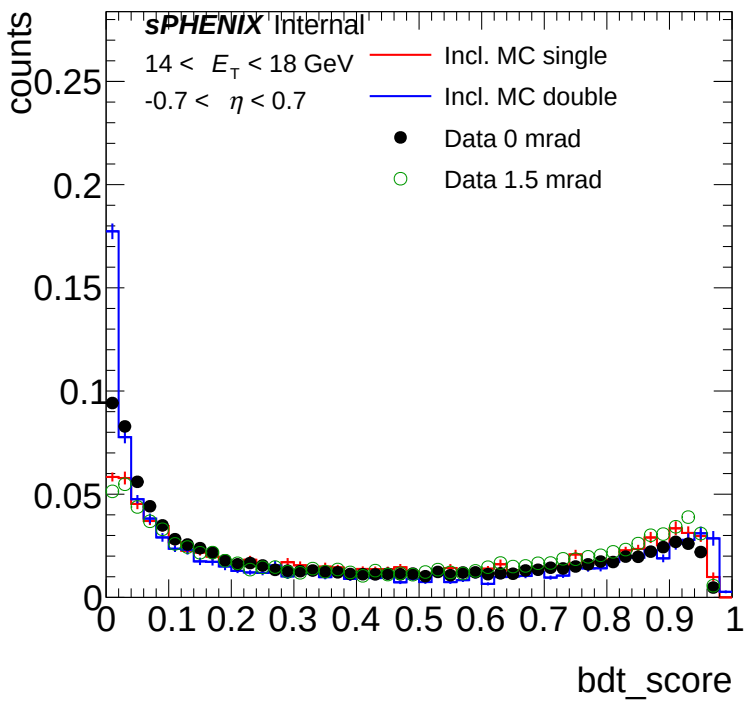


Figure 14: BDT score, $14 < E_T < 18\,\mathrm{GeV}$ (pt1). Double MC is pushed toward lower (more background-like) scores. Data 0 mrad sits closer to the double MC than data 1.5 mrad.

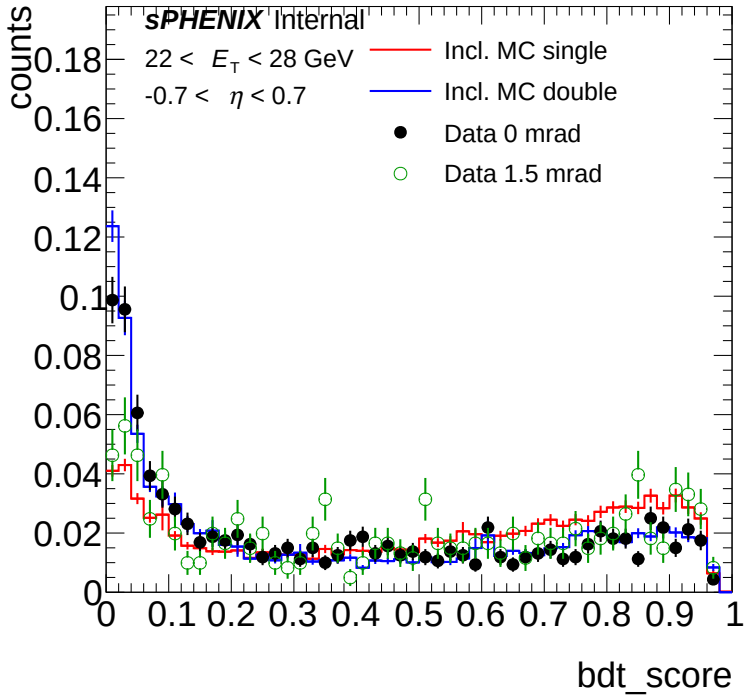


Figure 15: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3).

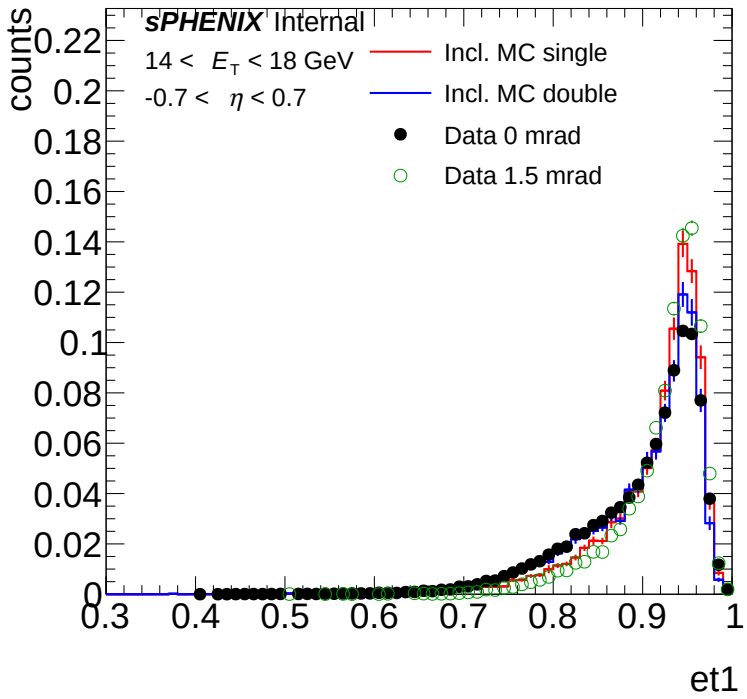


Figure 16: f_1, $14 < E_T < 18\,\text{GeV}$ (pt1). Double MC shifts f_1 lower as energy redistributes from the core to outer rings.

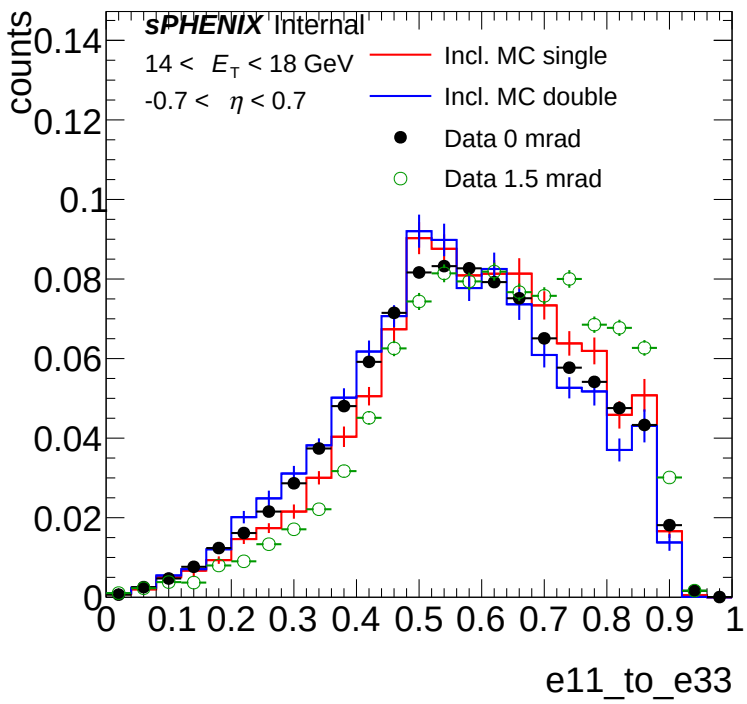


Figure 17: $E_{1\times1}/E_{3\times3}$, $14 < E_T < 18\,\text{GeV}$ (pt1). The single–double separation is smaller for this variable, consistent with the modest χ^2/ndf improvement seen in Sec. 2.4.

The crossing-angle dependence is clearly visible: data at 0 mrad (18.7% double-interaction rate) systematically sits closer to the double MC than data at 1.5 mrad (7.2%), as expected from the different pileup fractions.

4 χ^2/ndf Summary

Table 1 compiles the Data–Inclusive MC χ^2/ndf values read from the plots for the four variables at selected E_T bins (0 mrad, cut1).

Table 1: Data–Inclusive MC χ^2/ndf for nominal (single-only) and mixed (18.7% double-blended) MC at cut1, 0 mrad. Lower is better.

Variable	E_T bin [GeV]	Nominal	Mixed	Improvement factor
w_η^{CoG}	10–14 (pt0)	47.37	29.08	1.6×
w_η^{CoG}	14–18 (pt1)	17.63	2.99	5.9×
w_η^{CoG}	22–28 (pt3)	16.88	0.75	22.5×
BDT score	14–18 (pt1)	12.18	1.63	7.5×
f_1	14–18 (pt1)	9.33	1.93	4.8×
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.51	4.14	1.1×

5 Discussion

The mixed MC dramatically improves data–MC agreement for the pileup-sensitive variables (w_η^{CoG} and BDT score), with χ^2/ndf improvement factors ranging from 1.6× to 22.5×. The

improvement grows with E_T: at 22–28 GeV, w_η^{CoG} χ^2/ndf drops from 16.88 to 0.75, indicating that the nominal MC's failure to model pileup-induced broadening becomes more significant relative to statistical uncertainties at higher E_T.

The energy fraction f_1 also benefits substantially (4.8× at pt1), consistent with double interactions redistributing energy from the shower core to outer rings. In contrast, $E_{1\times1}/E_{3\times3}$ shows only a 1.1× improvement at pt1, suggesting this ratio is less sensitive to the pileup-induced broadening or that competing effects partially cancel.

The effect is largest at low E_T in absolute terms (highest raw χ^2/ndf values) but the relative improvement factor grows with E_T, where the pileup fraction is highest relative to the cluster energy and the nominal MC discrepancy is most systematic rather than statistical.

6 Conclusions

Blending 18.7% full-GEANT double-interaction MC with nominal single-interaction MC substantially improves data–MC shower shape agreement for pileup-sensitive variables, with χ^2/ndf improvement factors up to 22×. This confirms that double interactions are a dominant source of data–MC discrepancy in w_η^{CoG} and BDT score distributions. The mixed MC should be considered as the baseline for the 0 mrad crossing-angle period, with residual data–MC differences absorbed into systematic uncertainties.